

Sit-out fairness: Option A implemented

Pickleball.ai scheduler — sit-outs are now compared as rates, not totals. Both findings re-measured.
15 August 2026.

What changed

The penalty used to level **lifetime sit-out totals**, with a newcomer starting at zero. It now levels **sit-outs per session attended**.

- **A player's own rate** is used once they have three or more recorded sessions.
- **Below three sessions — including none at all** — the section average is used instead. One or two nights is noise, and it would otherwise dominate the scoring.
- **No history for the section** leaves the penalty skipped entirely, exactly as before.

One design correction was needed along the way. Dividing each player by their *own* attendance still penalised the newcomer, because a short record is dragged further by a single night than a long one. Every player is now projected over the same horizon, so only the rates are compared:

```
function rateIncludingTonight(baseline, sitOutsTonight) {
  const sitOutsOverHorizon =
    baseline.rate * HISTORY_RATE_HORIZON_NIGHTS
    + sitOutsTonight;
  return sitOutsOverHorizon / (HISTORY_RATE_HORIZON_NIGHTS + 1);
}
```

How the numbers were measured

Five players, one court, four rounds — four bench places to share among five people, so exactly one player escapes the bench each night and history decides who. The old raw-count scoring was replayed through the same harness for a like-for-like comparison. Finding 1 was run 300 times (1,200 bench places, an even share being 240 each); Finding 2 twenty times.

Finding 1 — the newcomer

	Before (totals)	After (rates)	Fair share
Newcomer, first night	300 (25.0%)	238 (19.8%)	240 (20%)
Four regulars combined	900 (75.0%)	962 (80.2%)	960 (80%)

The newcomer lands on her fair share. Her starting rate is the section average of 6.0 per night rather than zero, so she arrives owing nothing and owed nothing. Before, she was benched in every single one of the 300 runs; now she is indistinguishable from the regulars, the remaining spread being ordinary chance (one standard deviation is about 7).

Finding 2 — Faithful against Occasional

	Rate used	Benched before	Benched after
Faithful (8 nights, 8 sit-outs)	1.00	17	20
Occasional (6 sit-outs)	1.12	20	0

The inversion is gone: Occasional is now benched *less* than Faithful, where before she was benched more despite having the worse deal every night she came.

Note the rate she is judged on. With only two recorded sessions she falls under the three-session guard, so the scheduler uses the section average of 1.12 rather than her own 3.0. That is the guard doing its job — and 1.12 is still above Faithful's 1.00, so she is favoured anyway. Given a third session, her own rate of 3.0 is used and the result is the same.

The correction stops when the debt is paid

A fair scheduler should repay Occasional and then stop, not favour her forever. Letting her attend every week from then on:

Week	1	2	3	4	5	6
Her rate	1.12	2.00	1.50	1.20	1.00	1.00
Sat out	0	0	0	0	1	1

Four weeks of protected play, then her rate meets the group's 1.00 and she rejoins the ordinary rotation. Week 2 rises to 2.00 because her third session unlocks her own rate; it then falls as she plays.

A bug the review caught

A sub-agent reviewing the change found that the switch turning this penalty off was testing the wrong thing. It asked “is there any saved history?” when what matters is “do the players’ rates actually differ?”. Where every rate is the same, the penalty measures nothing but tonight’s imbalance, which is already charged for elsewhere — so it was billed twice.

This was not a rare edge case. With the three-session guard, nobody can be judged on their own rate until a section has three saved nights, so **every section’s first two nights hit it, guaranteed**, as did any night where none of the players present appear in the record. The effect was to inflate the ordinary sit-out penalty from 8 to 14 per unit — a 75% distortion against every other consideration in the score, carrying no information at all.

Case	History term before fix	After fix
Section with 2 saved nights	6.0 points of noise	0
None of tonight's players in the record	6.0 points of noise	0
Group who genuinely all share a rate	6.0 points of noise	0
Rates that genuinely differ	138 points	138 points

The last row is the check that matters in the other direction: the fix silences the penalty only where it was saying nothing, and leaves it at full strength where it has something to say.

What deliberately did not change

- A section with no saved history scores exactly as before — the history sit-out term stays at zero, so tonight's imbalance is not counted twice.
- History saved against a different section is ignored, as before.
- Saving a session is still a deliberate click. Nothing auto-saves.
- Repeated partners and opponents from past sessions are untouched.
- Backup files need no migration: rates are worked out at generate time from what is already saved, so nothing new is written to disk.

Both findings from the original analysis are resolved. The change is in index.html and running.